Time Frequency Analysis

Window Functions

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Short Time Fourier Transform

- **Idea: analyze incoming signal to detect events and when they occurred**
  - Events: frequencies
  - When: time
- ***s[n]*: *L* samples from *x[n]***
- **Perform FFT on the segment *s[n]* of input *x[n]***
- **How to obtain *s[n]*?**
  - *s[n] = x[n]*w[n]*
  - *w[n]: a* window function

Types of Window

The following window functions *w[n]* will be introduced:

- Rectangular window
- Hamming window
- Hanning/Hann window
- Bartlett/Triangular window

Assume *w[n]* is of length *L*.

Rectangular Window

Causal:

$$w_R[n] = \begin{cases} 1, & 0 \le n \le L-1 \\ 0, & o/w \end{cases}$$

Symmetric:

$$w_{R,sym}[n] = \begin{cases} 1, & -\frac{L-1}{2} \le n \le \frac{L-1}{2} \\ 0, & o/w \end{cases}$$

Relationship:

- $w_R[n] = w_{R,sym}\left[n - \frac{L-1}{2}\right]$
- $W_R(e^{j\omega}) = W_{R,sym}(e^{j\omega})e^{-j\omega(\frac{L-1}{2})}$ (time shifting property of DTFT)
- $|W_R(e^{j\omega})| = |W_{R,sym}(e^{j\omega})|$, i.e. same amplitude response

What is $W_{R,sym}(e^{j\omega})$?

- $w_{R,sym}[n] = \begin{cases} 1, & -\frac{L-1}{2} \le n \le \frac{L-1}{2} \\ 0, & o/w \end{cases}$

- $W_{R,sym}(e^{j\omega}) = \sum_{-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=-\frac{L-1}{2}}^{\frac{L-1}{2}} e^{-j\omega n} = \frac{e^{j\omega\frac{L-1}{2}}(1-e^{-j\omega L})}{1-e^{-j\omega}}$

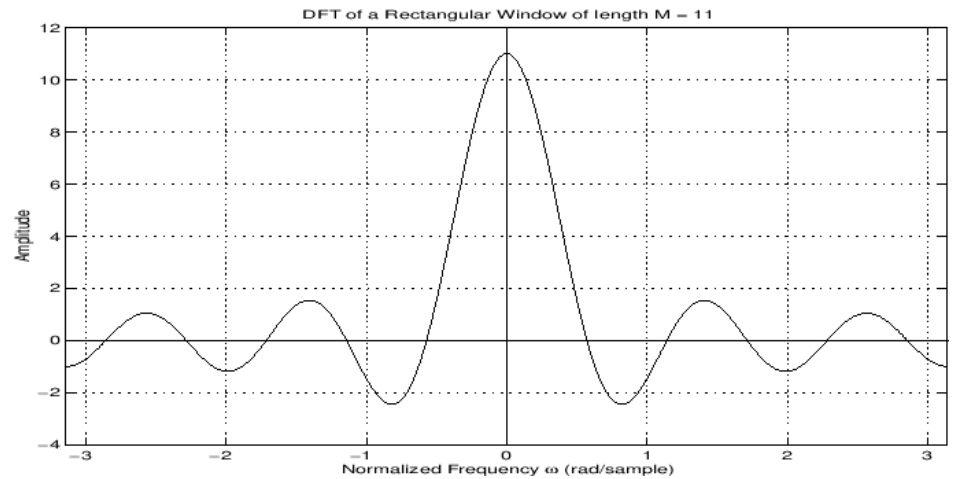
$$= \frac{e^{-j\omega\frac{1}{2}}e^{j\omega\frac{L}{2}}(1-e^{-j\omega L})}{1-e^{-j\omega}} = \frac{(e^{j\omega\frac{L}{2}} - e^{-j\omega\frac{L}{2}})}{e^{j\omega\frac{1}{2}}(1-e^{-j\omega})} = \frac{(e^{j\omega\frac{L}{2}} - e^{-j\omega\frac{L}{2}})}{(e^{j\omega\frac{1}{2}} - e^{-j\omega\frac{1}{2}})} = \frac{sin(\frac{\omega L}{2})}{sin(\frac{\omega}{2})}$$

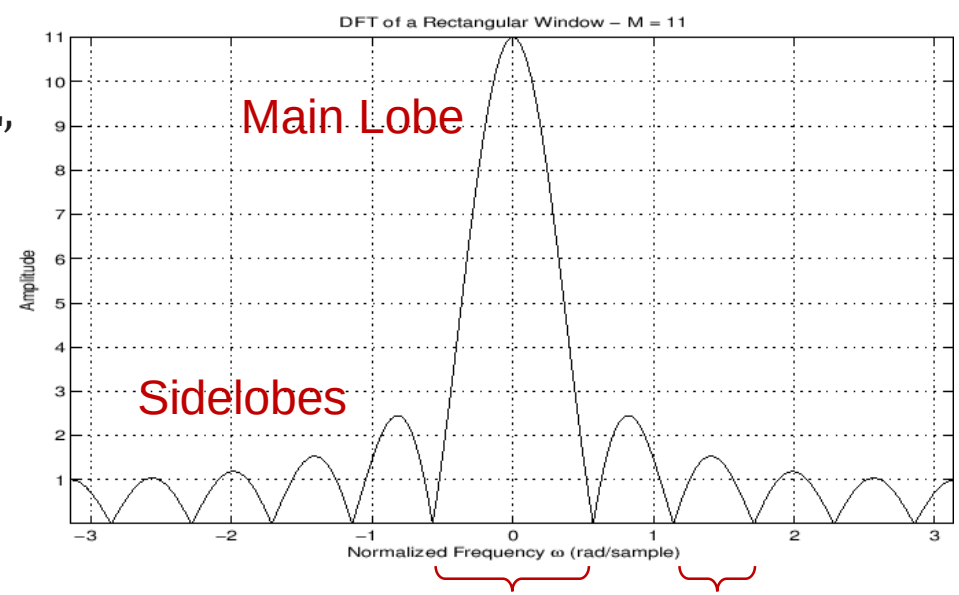
$$= L * asinc_L(\omega)$$

- **Aliased sinc function:** $asinc_L(\omega) = \frac{sin(L\frac{\omega}{2})}{Lsin(\frac{\omega}{2})}$

Amplitude Characteristics

- $\left|W_{R,sym}\left(e^{j\omega}\right)\right| = \left|\frac{sin(\frac{\omega L}{2})}{sin(\frac{\omega}{2})}\right|$
- **$\omega = 0$:** $\left|W_{R,sym}\left(e^{j\omega}\right)\right| = L$
- **Zero crossings:** $\frac{\omega L}{2} = k\pi, \omega = \frac{2k\pi}{L}$, **k is a non-zero integer**
  - i.e., $\omega = \ldots, -4\pi/L, -2\pi/L, 2\pi/L, 4\pi/L, \ldots$
- **Main lobe width: 4π/L**
- **Zeros spaced uniformly (except main lobe): 2π/L**

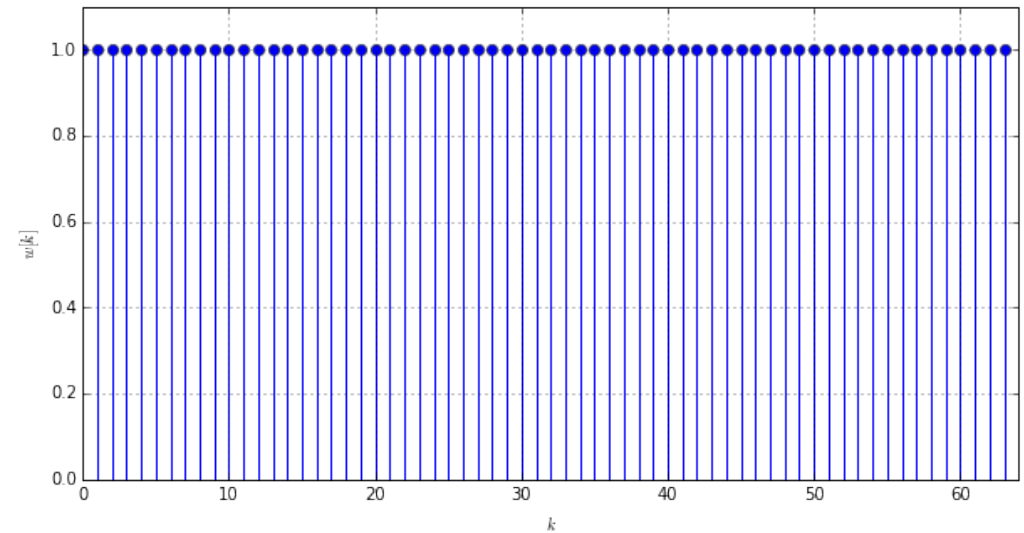


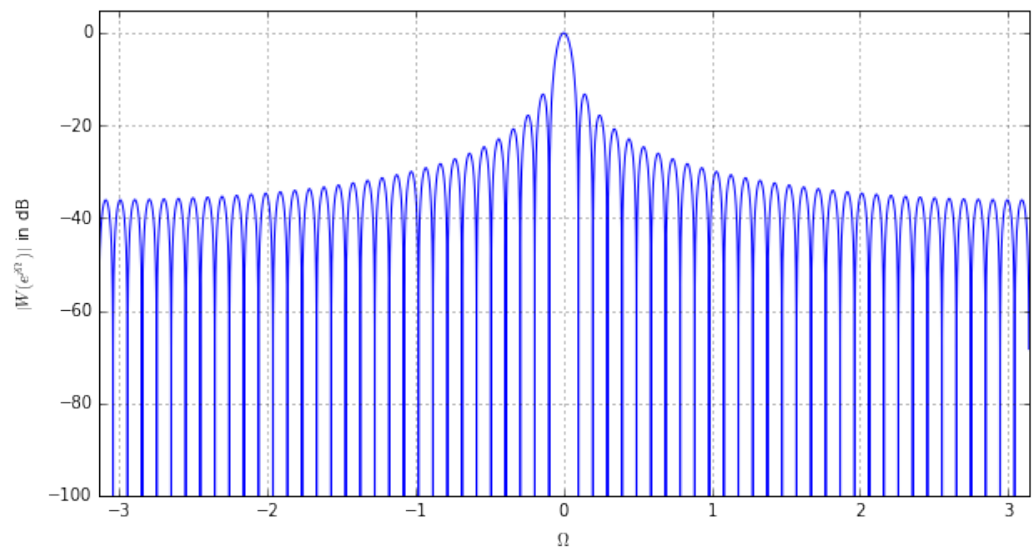


Images from: https://ccrma.stanford.edu/~jos/Intro421/Rectangular_Window_Transform_Cont_d.html

Rectangular Window

- $L = 64$
- $w_R[n] = 1,$
- $0 \leq n \leq L - 1$





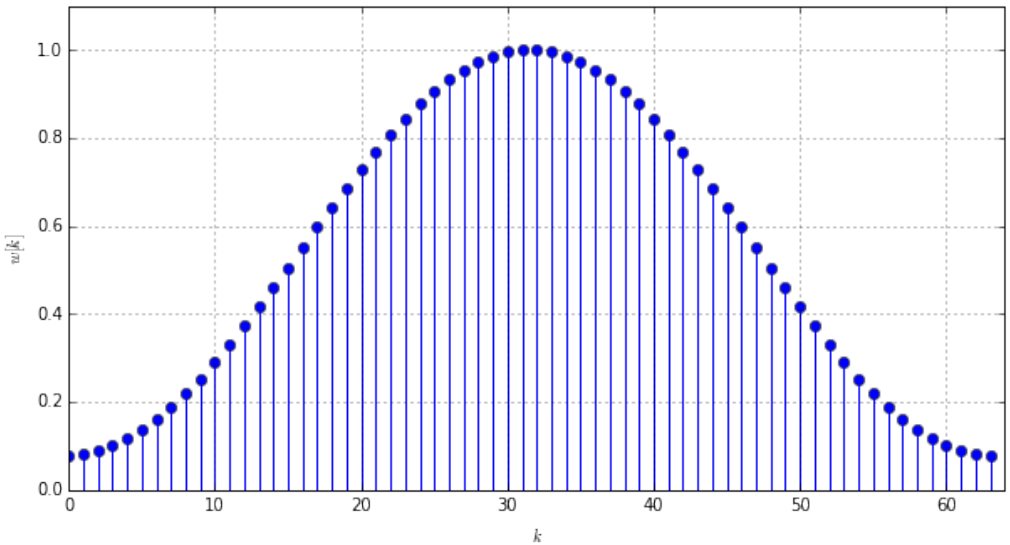
Hamming Window

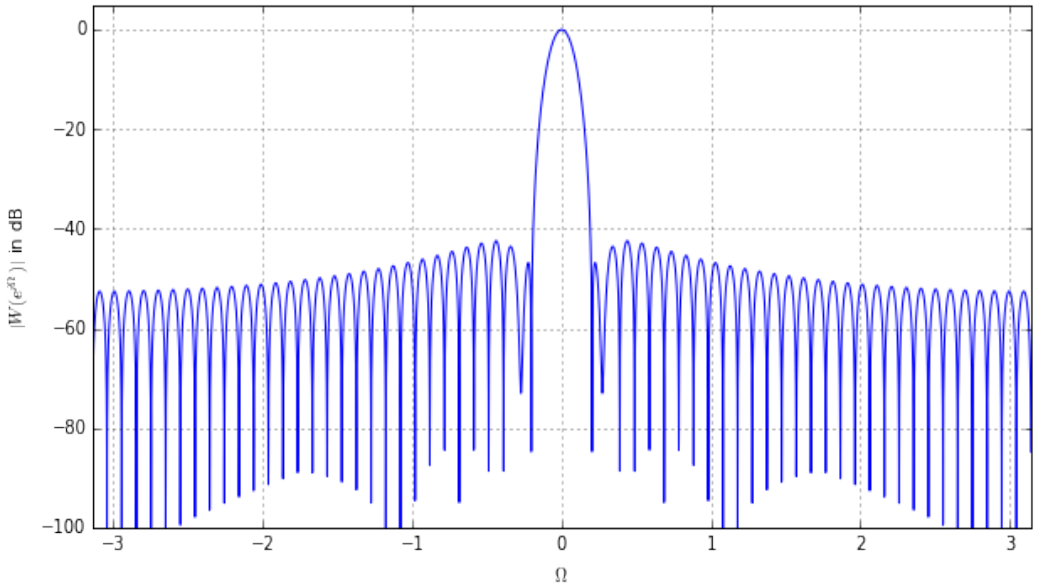
$$w_{Ham}[n] = \begin{cases} 0.54 - 0.46\cos(\frac{2\pi n}{L-1}), & 0 \le n \le L-1 \\ 0, & o/w \end{cases}$$

- $w_{Ham}[0] \neq 0, w_{Ham}[L-1] \neq 0$
- $W_{Ham}(e^{j\omega})$: **"sinc" function form**
  - Main lobe width = 8π/L radians (twice of the rectangular window)
  - Smaller side lobes

Hamming Window

- $L = 64$
- $w_{Ham}[n] = 0.54 - 0.46cos\left(\frac{2\pi n}{L-1}\right),$
- $0 \leq n \leq L-1$





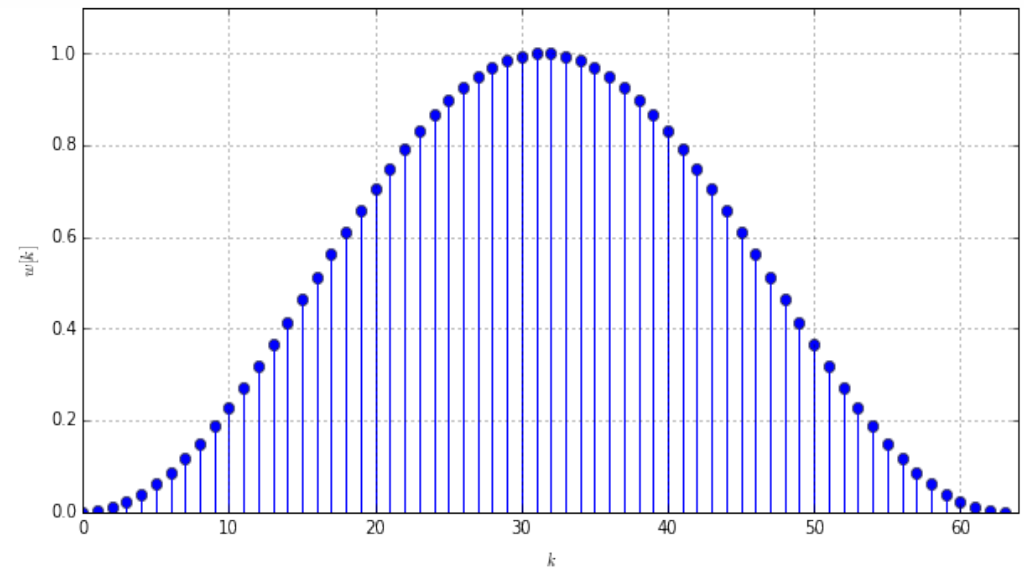
Hanning/Hann Window

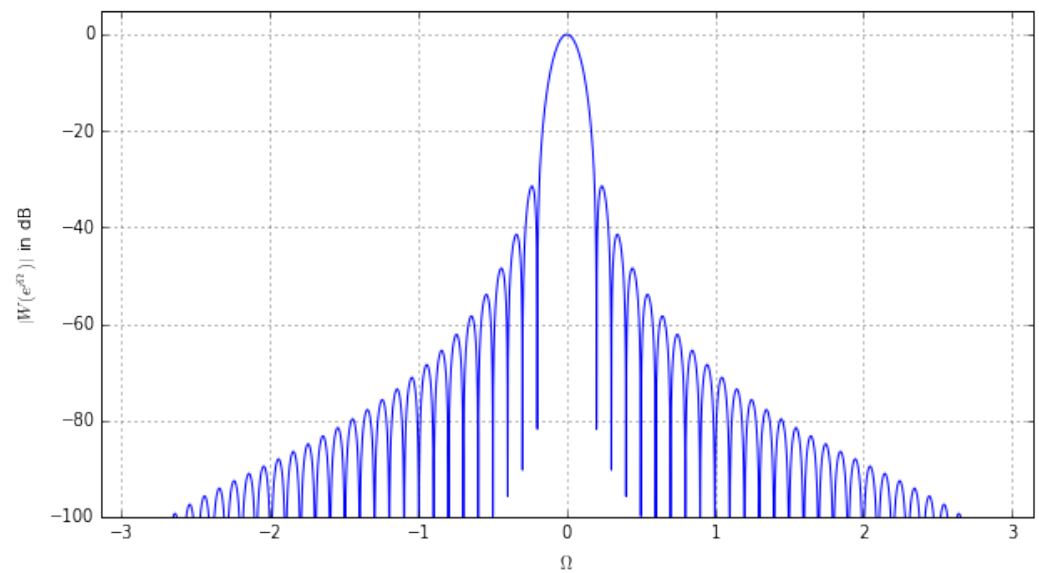
$$w_{Han}[n] = \begin{cases} 0.5 - 0.5\cos(\frac{2\pi n}{L-1}), & 0 \le n \le L-1 \\ 0, & o/w \end{cases}$$

- $w_{Han}[0] = 0, w_{Han}[L-1] = 0$
- $W_{Han}(e^{j\omega})$: **"sinc" function form**
  - Main lobe width = 8π/L radians (twice of the rectangular window)
  - Fast decay of side lobes

Hanning/Hann Window

- $L = 64$
- $w_{Han}[n] = 0.5 - 0.5\cos\left(\frac{2\pi n}{L-1}\right),$
- $0 \leq n \leq L-1$



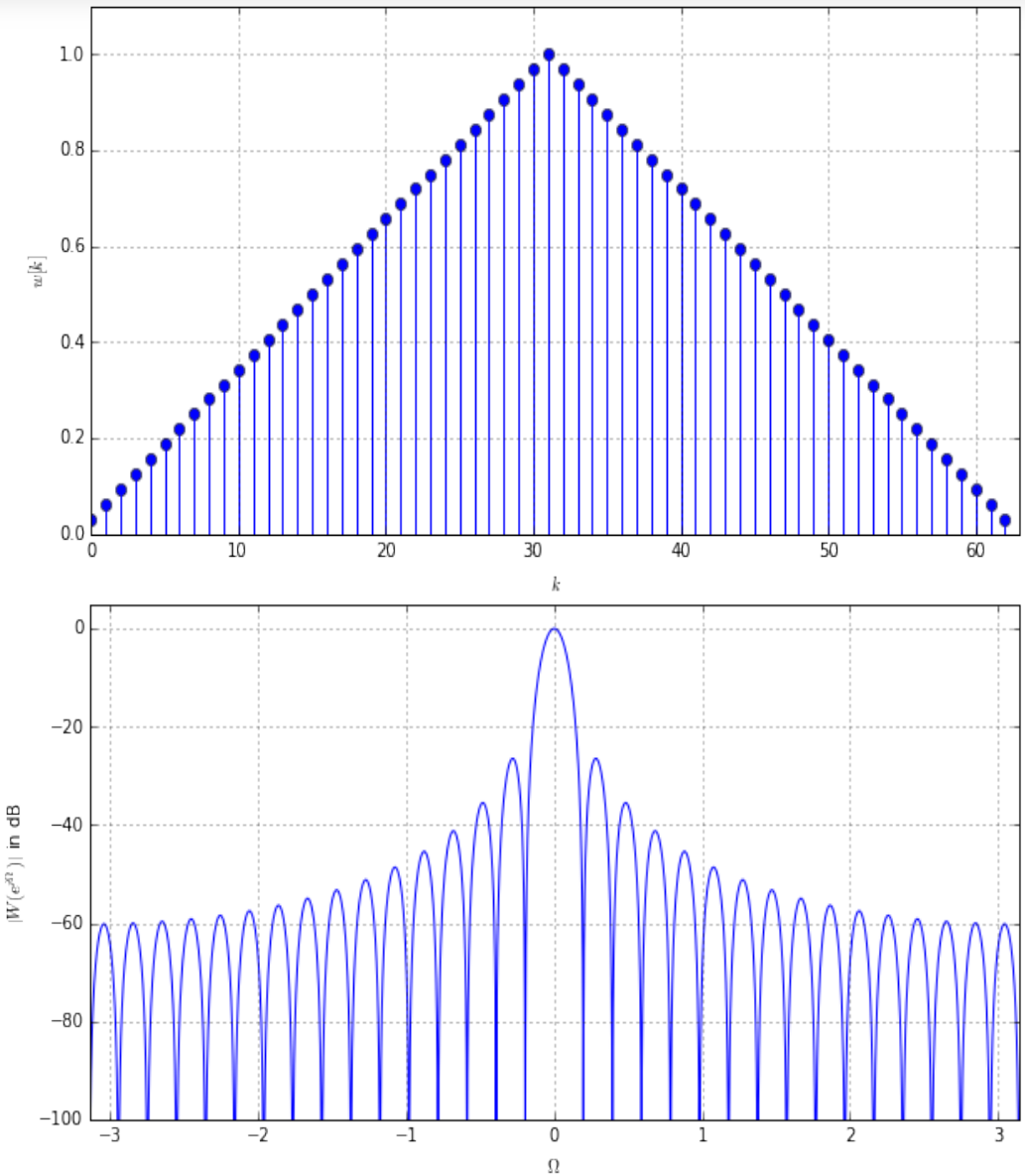


Bartlett/Triangular Window

- $w_{Bart}[n] = w_R[n] * w_R[n]$
  - It is the convolution of two rectangular windows $w_R[n]$.
  - $w_R[n]$: length (L+1)/2
  - $w_{Bart}[n]$: length (L+1)/2*2-1 = L
- $W_{Bart}(e^{j\omega}) = W_R(e^{j\omega})W_R(e^{j\omega}) = |W_R(e^{j\omega})|^2 e^{j2\angle W_R(e^{j\omega})}$
  - Main lobe width = 4π/((L+1)/2) = 8π/(L+1) radians (almost twice of the rectangular window 4π/L)
  - Fast decay of side lobes

Bartlett/Triangular Window

- $L = 63$
- $w_{Bart}[n]$



Question

Which window function has the smallest main lobe width in the frequency domain?

a) Rectangular window

b) Hamming window

c) Hanning window

d) Triangular window

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